IB MATHEMATICS SL • UNIT 1.01

C Topic 1: Number & Algebra (Sequences, Logs & Binomial)

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Topic 1: Number & Algebra (Sequences, Logs & Binomial) Practice Exercises

1 MCQ

Consider the function $f(x) = 2x + 5$. What is the value of $f(3)$?

A 11
B 8
C 10
D 16

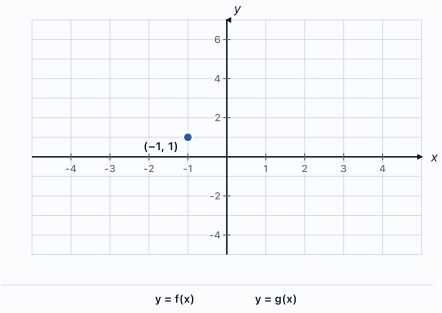
2 MCQ

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown below. Find the value of the y-intercept.

A 5
B -5
C 0
D 1

3 MCQ

Consider the function $f(x) = x^2 - 4x + 3$ and the line $g(x) = x - 1$. The graph of these functions is shown below. Find the coordinates of the point where the line intersects the parabola.



A (1, 0)
B (3, 2)
C (2, 1)
D (4, 3)

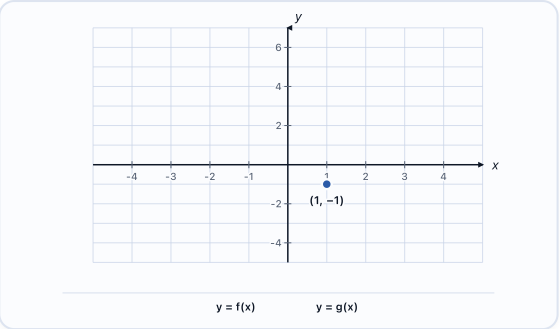
4 MCQ

$e^0 =$

A 0
B 1
C e
D ∞

5 MCQ

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. It passes through the points $(0, -3)$, $(1, 0)$, and $(-1, -4)$. Find the value of c.



A -3

B 3

C -1

D 1

6 MCQ

Consider the function $f(x) = 2x - 3$. Determine the value of $f(3)$.

A 3

B 6

C 9

D 0

7 MCQ

Solve for x: $3^x = 9$.

A 3

B 4

C 2

D 1

8 MCQ

An arithmetic sequence has first term $u_1 = 7$ and common difference $d = 2$. Find u_{15}.

A 37

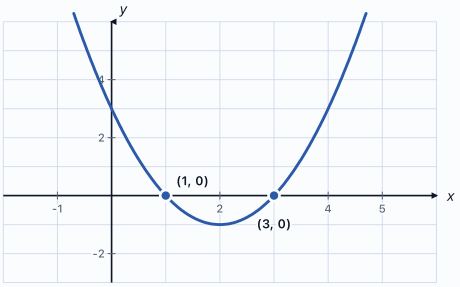
B 35

C 33

D -21

9 MCQ

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. What are the coordinates of the x-intercepts of the function?



A (0, 1) and (0, 3)

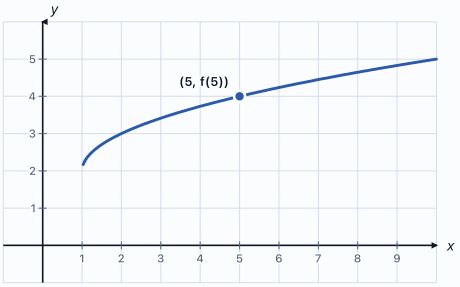
B (1, 0) and (3, 0)

C (-1, 0) and (-3, 0)

D (0, -1) and (0, -3)

10 MCQ

The graph of the function $f(x) = \sqrt{x-1} + 2$ is shown below. What is the value of $f(5)$?



A 2

B 4

C 3

D 5

11 MCQ

An arithmetic sequence has first term $u_1 = 4$ and common difference $d = -1$. Find u_{15}.

A -11

B 18

C -10

D -9

12 MCQ

Evaluate $\log_{10}(10000)$.

A 2

B 3

C 4

D 1

13 MCQ

A geometric sequence has first term $u_1 = -4$ and common ratio $r = -3$. Find u_7.

A 972

B -2912

C 8748

D -2916

14 MCQ

Given $\log_5(6) = p$, express $\log_5(6^3)$ in terms of p.

A $\frac{p}{3}$

B p^3

C $3p$

D $p + 3$

15 MCQ

An arithmetic sequence has first term $u_1 = 4$ and common difference $d = -4$. Find u_{19}.

A 76

B -68

C -72

D -64

16 MCQ

A geometric sequence has first term $u_1 = 5$ and common ratio $r = -3$. Find u_6.

A 3645

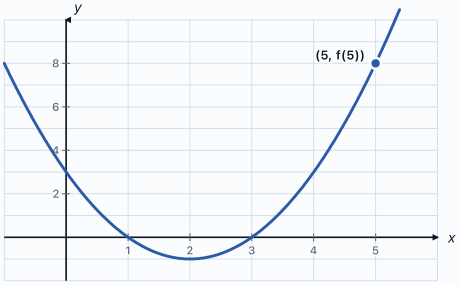
B -1215

C 405

D -1216

17 MCQ

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. Find the value of $f(5)$.



A 12

B 18

C 20

D 8

18 MCQ

Determine the numerical value of $\log_5(125)$.

A 1

B 3

C 4

D 2

19 MCQ

An arithmetic sequence has first term $u_1 = 3$ and common difference $d = 6$. Find u_{19}.

A 117

B 105

C -105

D 111

20 MCQ

A geometric sequence has first term $u_1 = -2$ and common ratio $r = \frac{1}{3}$. Find u_6.

A 1

B $-\frac{2}{243}$

C $-\frac{2}{729}$

D $-\frac{2}{81}$

21 MCQ

A geometric sequence has first term $u_1 = -1$ and common ratio $r = -3$. Find u_5.

A 27

B -91

C -81

D 243

22 MCQ

Find the root(s) satisfying for x: $5^x = 15625$.

A 3

B 6

C 2

D 4

23 MCQ

Calculate the solution set for for x: $3^x = 81$.

A 3

B 1

C 2

D 4

24 MCQ

A geometric sequence has first term $u_1 = -1$ and common ratio $r = 3$. Find u_6.

A -729

B -243

C -253

D -81

25 MCQ

A geometric sequence has first term $u_1 = -2$ and common ratio $r = 2$. Find u_5.

A -16

B -31

C -64

D -32